

W207 Final Project: NLP Sentiment Analysis

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Research Question

*How does the **accuracy** of **sentiment analysis** for company reviews* vary across machine learning models?*

** customer feedback*

Background

Why Do Company Reviews Matter?

Understanding customer feedback allows us to extract actionable insights from large datasets.

Real Work Applications

- **Amazon:** 35% of revenue is driven by feedback-based recommendation engine.
- **Netflix:** Feedback informs trend predictions, helps tailor recommendations, and optimizes personalized marketing.
- **Coca Cola:** Sentiment analysis run on online mentions/feedback to measure brand loyalty.

Impact

Optimized user experience (higher customer satisfaction) and increased financial growth.

Key Insight

Companies that leverage customer analytics extensively are **2x** as likely to generate above-average profits and sales growth (*McKinsey, 2013*).

Existing Company Review Studies

"Customer Sentiment Analysis Through Social Media Feedback: A Case Study On Telecommunication Company"

- SVM outperformed Random Forest and Naïve Bayes for Twitter customer data

Sentiment Analysis of Customer Feedback and Reviews for Airline Services using Language Representation Model

- Importance of multimodal data in sentiment analysis

"Sentiment Analysis of Yelp Reviews: A Comparison of Techniques and Models"

- Logistic Regression, SVM, LSTM, and BERT on over 350,000 Yelp reviews – (logistic regression had 91% accuracy, outperforming complex models)

"A Systematic Literature Review on Machine Learning Applications for Consumer Sentiment Analysis Using Online Reviews"

- Naive Bayes and SVM achieved 88%-92% accuracy for sentiment classification

Our Approach

- Evaluate the performance of various models
 - Train models on **24,000** company reviews (e.g. customer feedback of businesses or services)
- Achieve high accuracy in sentiment analysis
 - Apply to other company reviews or other review applications
 - Use models to extract insights and analyze sentiment from text data

Results Summary

Model	Train Accuracy	Train Loss	Test Accuracy	Test Loss
Baseline (Logistic Regression Model)	0.967	0.111	0.663	0.635
Random Forest	0.965	0.192	0.801	0.468
LSTM	0.985	0.046	0.952	0.161
Bidirectional LSTM	0.977	0.069	0.963	0.111
CNN	0.968	0.097	0.961	0.128

Data

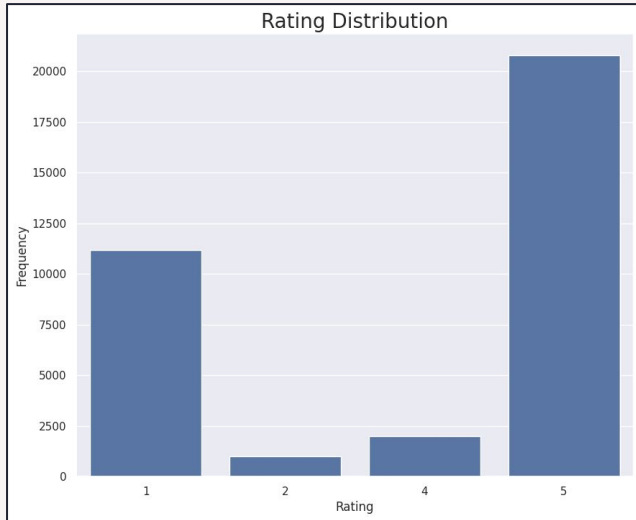
Our dataset consists of **100,000** company reviews extracted from **45** different companies.

The columns are split as follows:

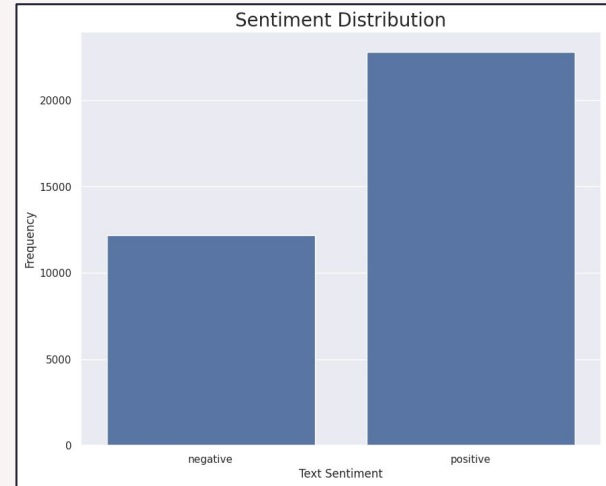
- **Id:** Unique review identifier
- **Review:** Customer review text
- **Rating:** Customer rating between 1-5

Data Preprocessing

The neutral reviews were dropped from the dataset to avoid training with ambiguous reviews and establish a clear distinction between positive and negative sentiment for the binary classification model.



12,000 positive & negative reviews were selected to create a balanced dataset for training



Data Preprocessing

Additionally, the preprocessing steps were taken:

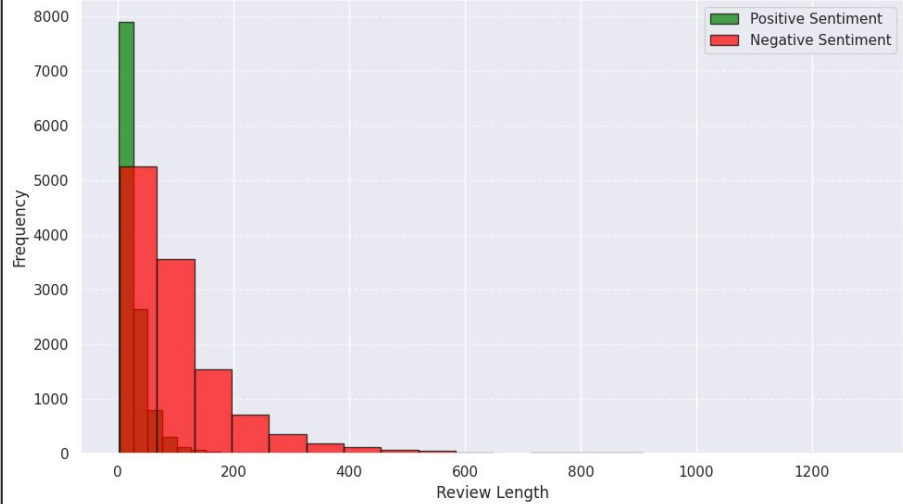
- Data was shuffled and split into train, test, and validation data.
- Punctuation and special (non-letter) characters were removed from the data.
- The y variable was converted from categorical (positive/negative) to binary (1,0).

Exploratory Data Analysis

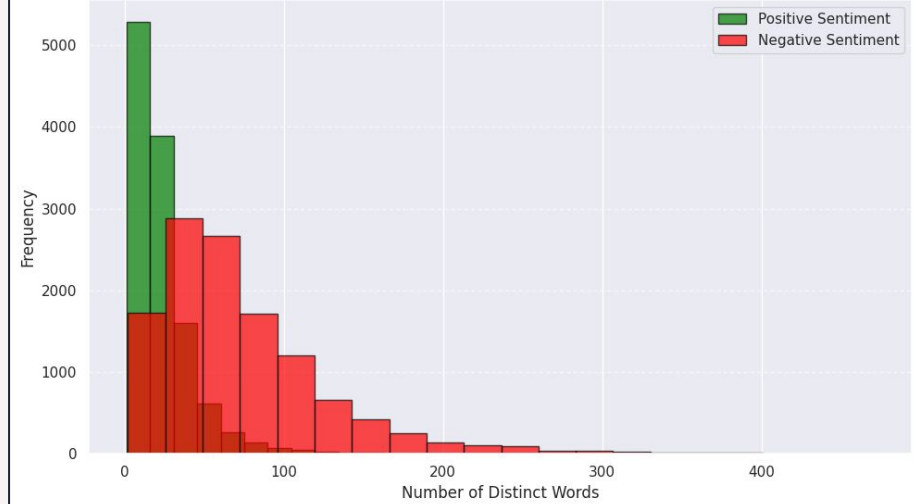
- The average positive review is 28 words long while the average negative review is 110 words long.
- Most reviews are short, with both sentiments having a lot of reviews at around the 0-100 word count range.
- Negative reviews dominate longer texts.

Exploratory Data Analysis

Distribution of Review Lengths by Sentiment



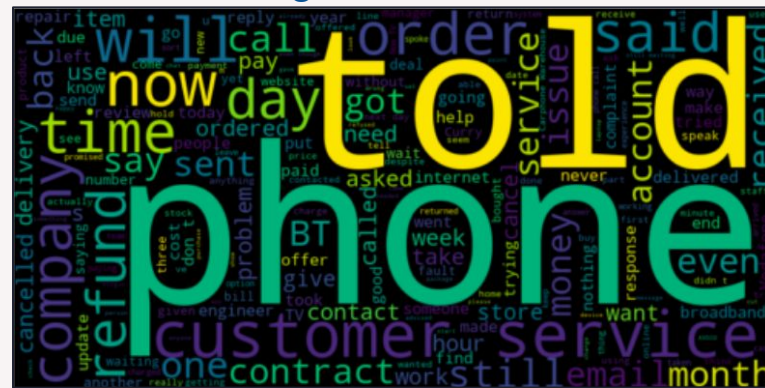
Distribution of Number of Distinct Words in the Reviews



Positive Reviews



Negative Reviews



Baseline Model

Logistic Regression

Definition: Predict probabilities for classification tasks, outputting results between 0 and 1.

Performance:

1. Training Accuracy: 0.967
2. Validation Accuracy: 0.679
3. Test Accuracy: 0.663

Classification Report:					
	precision	recall	f1-score	support	
0	0.51	0.73	0.60	4088	
1	0.81	0.63	0.71	7577	
accuracy			0.66	11665	
macro avg	0.66	0.68	0.65	11665	
weighted avg	0.71	0.66	0.67	11665	

Model 1: Random Forest

Definition: Algorithm that uses decision trees to make predictions, combining their results to improve accuracy.

Model Parameters:

1. Number of trees: 40
2. Maximum depth: 20

Performance:

1. Training Accuracy: 0.9647
2. Validation Accuracy: 0.7623
3. Test Accuracy: 0.8009

Random Forest - Classification Report:					
	precision	recall	f1-score	support	
0	0.75	0.64	0.69	4088	
1	0.82	0.89	0.85	7577	
accuracy			0.80	11665	
macro avg	0.79	0.76	0.77	11665	
weighted avg	0.80	0.80	0.80	11665	

Model 2: Long Short-Term Memory (LSTM)

Definition: Neural network that processes sequences of data by remembering important information over time.

Model Parameters:

1. Maximum Vocab Size: 10000
2. Maximum Sequence Length: 50
3. Embedding Dim: 100

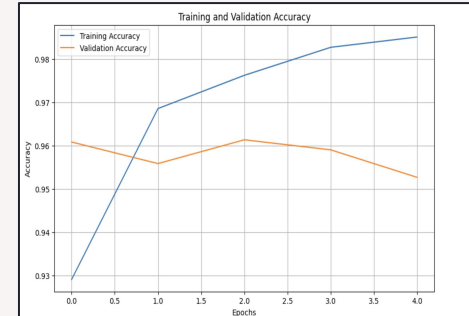
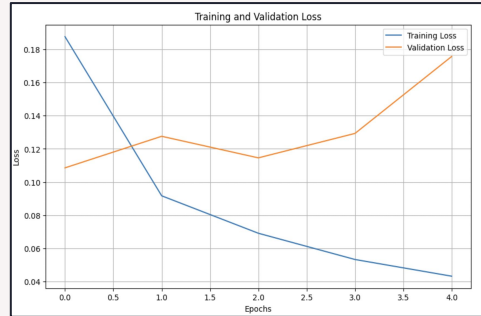
Performance:

1. Training Accuracy: 0.985
2. Test Accuracy: 0.951
3. Validation Accuracy: 0.954
4. Train Loss: 0.046
5. Test Loss: 0.161
6. Validation Loss: 0.153

Model: "sequential_8"

Layer (type)	Output Shape	Param #
embedding_5 (Embedding)	(32, 50, 100)	1,000,000
lstm_6 (LSTM)	(32, 64)	42,240
dense_12 (Dense)	(32, 1)	65

Total params: 3,126,917 (11.93 MB)
Trainable params: 1,042,305 (3.98 MB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 2,084,612 (7.95 MB)



Model 3: Bidirectional LSTM

Definition: Neural network that processes sequential input in forward and backward direction to capture contextual info

Model Parameters:

1. Maximum Vocab Size: 1000
2. Maximum Sequence Length: 400
3. Embedding Dim: 10
4. Optimizer: Adam; Learning Rate = 0.001; Loss: Binary Cross Entropy

Performance:

1. Final Training Accuracy: 0.977
2. Final Validation Accuracy: 0.963
3. Test Accuracy: 0.963
4. Final Training Loss: 0.069
5. Final Validation Loss: 0.114
6. Test Loss: 0.111

Model Architecture:

Embedding

(input_dim=vocab_size,output_dim=embedding_dim,input_length=max_length)

Recurrent Layer 1

Bidirectional(LSTM(64, return_sequences=True, dropout=0.2))

Recurrent Layer 2

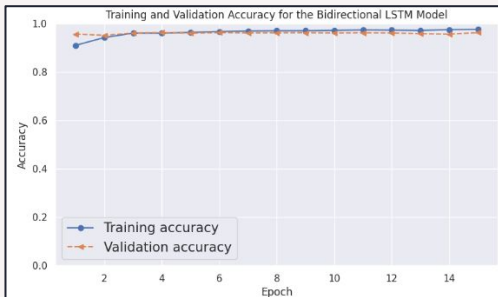
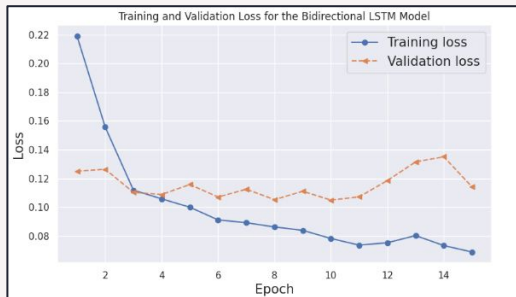
Bidirectional(LSTM(32, dropout=0.2))

Dense Layer

Dense(32, activation='relu')

Output

Final layer (binary classification): Dense(1, activation='sigmoid')

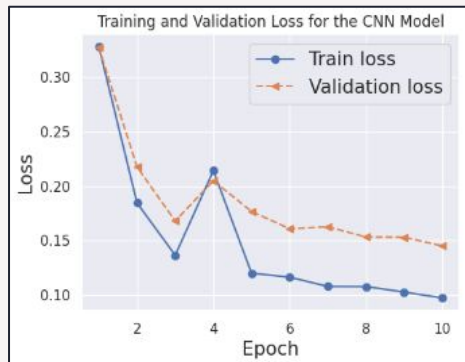


Model 4: CNN

Definition: Model that applies 1D convolutional layers to capture local patterns in the input

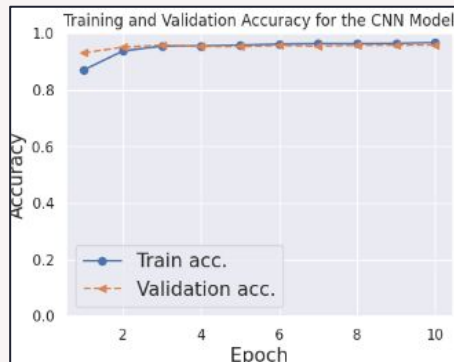
Model Parameters:

1. Maximum Vocab Size: 100
2. Maximum Sequence Length: 400
3. Embedding Dim: 10
4. Optimizer: Adam; Learning Rate = 0.001; Loss: Binary Cross Entropy



Performance:

1. Final Training Accuracy: 0.968
2. Final Validation Accuracy: 0.959
3. Test Accuracy: 0.961
4. Final Training Loss: 0.097
5. Final Validation Loss: 0.145
6. Test Loss: 0.1278



Model Architecture:

Embedding

(input_dim=vocab_size,output_dim=10,input_length=400)

Convolutional Layer 1

Conv1D(filters=128,kernel_size=4,padding='same',activation='relu')
Dropout(rate=0.5)
MaxPooling1D(pool_size=2)

Convolutional Layer 2

- Conv1D(filters=64,kernel_size=4,padding='same',activation='relu')
- Dropout(rate=0.5)
- MaxPooling1D(pool_size=2)
- Average the embeddings: GlobalAveragePooling1D()
- Dropout(rate=0.5)

Final output layer

- Dense(1, activation='sigmoid')

Experiments Summary

1

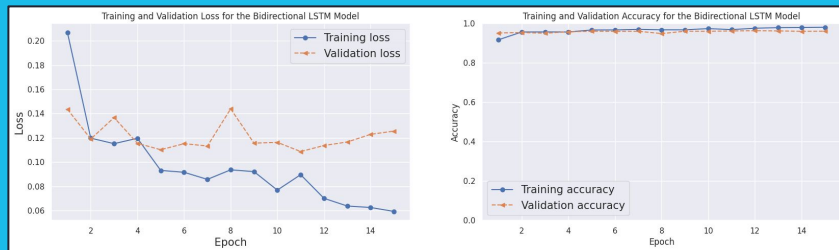
Increasing embeddings from 10 to 100 didn't affect accuracy scores based on results from LSTM and BiLSTM.

Removing the 3rd convolutional layer did not have an effect on final loss and accuracy that was notable.

2



Removing dropout from Bidirectional LSTM layers led to reduced training loss but increased validation loss after 10 epochs, suggesting potential overfitting.



3

Observation: Already high accuracy with simple models can mean hyperparameter tuning is not necessary.

Conclusions

Model	Train Accuracy	Train Loss	Test Accuracy	Test Loss
Baseline (Logistic Regression Model)	0.967	0.111	0.663	0.635
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Conclusions: What We Learned



Simple Model



Complex Model

Simple models risk overfitting by memorizing patterns, while LSTM architectures leverage complexity to capture nuanced relationships and contextual word dynamics.

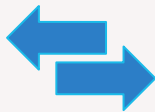


Random Forest underperformed as it lacks the ability to capture contextual relationships in text or handle sequential data, making it less effective for complex, unstructured inputs.

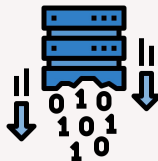
Conclusions: What We Learned



High baseline accuracy highlighted the model's strong performance, demonstrating that hyperparameter tuning may have diminishing returns in well-optimized setups.



Bidirectional LSTM outperforms standard LSTM with higher test accuracy and lower test loss, highlighting the advantage of leveraging both past and future contexts for more comprehensive feature extraction.

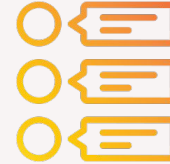


Low training loss in advanced architectures like LSTM and Bidirectional LSTM demonstrates that carefully designed models can achieve strong generalization, emphasizing the importance of selecting architectures that balance training and testing performance.

Conclusions: Future Opportunities



Running the model on other review applications (e.g. GlassDoor employee reviews)



Topic-based sentiment analysis



Extend the model to generate responses



Questions?

Code Submission: GitHub Link

https://github.com/sudikshas/sentiment_classification_project/tree/project

Contributions

Sudiksha Sarvepalli: EDA, Data Preprocessing, Model 3 (Bidirectional LSTM), Model 4 (CNN), Experiments Summary

Shruti Mahadevan: EDA, Baseline Model, Model 1 (Random Forest), Model 2 (LSTM)

Pauline Ranjan: EDA, Experiments Summary, Conclusions

Eduardo Jose Villasenor: Research Question, Our Approach, Data/Data Preprocessing/EDA/Model Slides Formatting

Akshita Gandra: Background, Existing Studies/Research, Our Approach, Test Data, Slide Formatting

Appendix

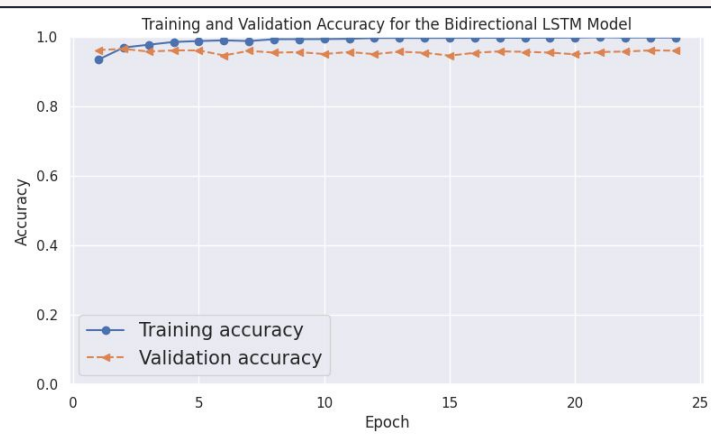
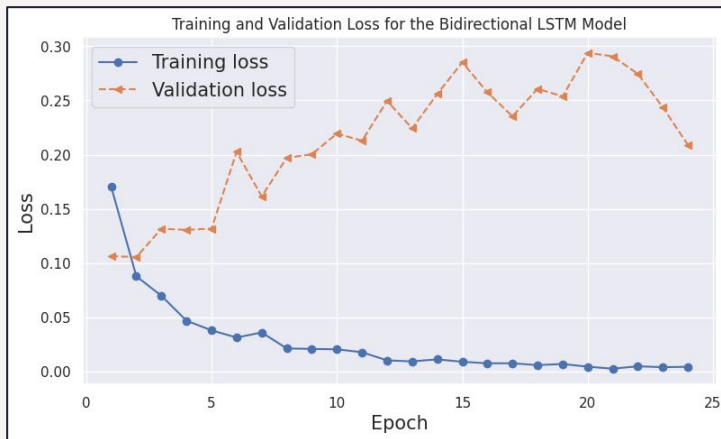
Model 3b: Bidirectional LSTM Experimentation

Model Hyperparameters:

1. Vocab size: 10000
2. Max Review Length: 200
3. Embedding dimension: 150
4. Optimizer: Adam with learning rate 0.001

Model Results:

1. Final Training Accuracy: 0.999
2. Final Validation Accuracy: 0.962
3. Final Training Loss: 0.004
4. Final Validation Loss: 0.209



Future Use Case (Employee Review) Studies

"Sentiment Analysis of Company Reviews Using Machine Learning"

- Employee reviews were analyzed using models like Random Forest and Gradient Boosting, achieving 85%-89% accuracy. The study emphasized the importance of feature extraction techniques like TF-IDF for improving model performance. (ieeexplore.ieee.org)

"Machine Learning for Company Review Sentiment Analysis"

- This study implemented NLP techniques with Gradient Boosting and BERT for analyzing employee sentiment, with BERT achieving a 94.5% accuracy. SHAP analysis revealed that negative sentiment was most influenced by terms related to workload and leadership issues. (link.springer.com)

Customer Feedback Studies (Notes)

"Sentiment Analysis of Yelp Reviews: A Comparison of Techniques and Models"

- This study compared Logistic Regression, SVM, LSTM, and BERT on over 350,000 Yelp reviews. Logistic Regression achieved 91% accuracy, outperforming more complex models like LSTM and BERT due to the data's simplicity and efficient feature engineering. (arxiv.org)

"A Systematic Literature Review on Machine Learning Applications for Consumer Sentiment Analysis Using Online Reviews"

- This study evaluated various ML models in the hospitality sector and found that Naive Bayes and SVM achieved 88%-92% accuracy for sentiment classification. Preprocessing steps like stemming and lemmatization were critical in achieving these results. (arxiv.org)

References (1/2)

- <https://www.kaggle.com/code/matthewjansen/nlp-sentiment-analysis-of-company-reviews#2-%7C-Data-Preprocessing>
- Background/Approach
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 - <https://www.intuz.com/blog/generative-ai-for-ecommerce-personalized-recommendation>
- Test Dataset ([W207 Test Data: Glassdoor Company Reviews](#))
 - https://www.glassdoor.com/Reviews/Integral-Ad-Science-Reviews-E782797_P3.htm?filter.iso3Language=eng
 - <https://www.glassdoor.com/Reviews/Lockheed-Martin-Reviews-E404.htm?filter.iso3Language=eng>
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 - <https://www.glassdoor.com/Reviews/Splunk-Reviews-E117313.htm>

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Thank You! :)